# Nu(Pr, Ra) Dataset: Sources and Provenance

Machine-learning prediction of the Nusselt number in 2D Rayleigh-Bénard convection

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October 17, 2026

## Overview

This document records exactly where every number in `data/nu_dataset_model.csv` came from. Section 3.3 of Shreshthi & Pandey's paper (*Turbulent Prandtl number in the near-wall region of thermal convection*) trains a random forest regressor to predict the Nusselt number $Nu$ from the Prandtl number $Pr$ and Rayleigh number $Ra$, using 55 data points drawn from five cited papers plus the paper's own simulations. This document reproduces the relevant tables from every source used, highlights which rows were kept, and comments on why the rest were left out. The dataset was later extended beyond the paper's five citations with four more papers that simulate the identical configuration (Sections 7–10), bringing it from 59 to 194 rows. Two of those sources report $Nu$ only in figures; their values were read from the vector graphics inside the PDFs (Section 14).

**Included** in `nu_dataset_model.csv` **Excluded** (see Comment column)

## References


1. Shreshthi and A. Pandey. *Turbulent Prandtl number in the near-wall region of thermal convection.* Project paper (`papers/shreshthi_pandey_2026_turbulent_prandtl_number.pdf`).

2. A. Pandey. *Thermal boundary layer structure in low-Prandtl-number turbulent convection.* J. Fluid Mech., 910:A13, 2021. doi:10.1017/jfm.2020.961 — arXiv:2101.03051.

3. A. Pandey and K. R. Sreenivasan. *Transient and steady convection in two dimensions.* J. Fluid Mech., 1015:A42, 2025. doi:10.1017/jfm.2025.10357 — arXiv:2503.03080.

4. A. Pandey, H. Tiwari, and K. R. Sreenivasan. *Thermal convection in 1, 2, 3, and 4 dimensions.* J. Fluid Mech., in press, 2026. arXiv:2607.26372.

5. E. P. van der Poel, R. J. A. M. Stevens, and D. Lohse. *Comparison between two- and three-dimensional Rayleigh-Bénard convection.* J. Fluid Mech., 736:177–194, 2013. doi:10.1017/jfm.2013.488. **Obtained.** The 2D $Nu$ values at $Ra = 10^8$ are tabulated in its Appendix Table 2 (not only plotted, as first assumed).

6. Y. Zhang and Q. Zhou. *Low-Prandtl-number effects on global and local statistics in two-dimensional Rayleigh-Bénard convection.* Phys. Fluids, 36(1):015107, 2024. doi:10.1063/5.0175011. **Obtained** (institutional access).

7. Y. Zhang, Q. Zhou, and C. Sun. *Statistics of kinetic and thermal energy dissipation rates in two-dimensional turbulent Rayleigh-Bénard convection.* J. Fluid Mech., 814:165–184, 2017. doi:10.1017/jfm.2017.19.

8. A. Xu, L. Shi, and H.-D. Xi. *Statistics of temperature and thermal energy dissipation rate in low-Prandtl number turbulent thermal convection.* Phys. Fluids, 31:125101, 2019. doi:10.1063/1.5129818 — arXiv:1911.04042.

9. J.-C. He, M.-W. Fang, Z.-Y. Gao, S.-D. Huang, and Y. Bao. *Effects of Prandtl number in two-dimensional turbulent convection.* Chin. Phys. B, 30(9):094701, 2021. doi:10.1088/1674-1056/ac0781.

10. E. P. van der Poel, R. J. A. M. Stevens, K. Sugiyama, and D. Lohse. *Flow states in two-dimensional Rayleigh-Bénard convection as a function of aspect-ratio and Rayleigh number.* Phys. Fluids, 24:085104, 2012. doi:10.1063/1.4744988 — arXiv:1206.3823.

11. J.-L. Yang, Y.-Z. Zhang, T.-C. Jin, Y.-H. Dong, B.-F. Wang, and Q. Zhou. *The Pr-dependence of the critical roughness height in two-dimensional turbulent Rayleigh-Bénard convection.* J. Fluid Mech., 911:A52, 2021. doi:10.1017/jfm.2020.1091. **Checked, not used** (Sec. 11).

Every included row describes the *same* physical setup: a closed 2D square box ($\Gamma = 1$), no-slip velocity on all four walls, isothermal top/bottom plates and adiabatic sidewalls. References 1–4 share one spectral-element code: several of their rows report the exact same $Nu$ to the decimal at the same $Pr, Ra$, which is only possible if it is the same simulation reused across papers. References 5–10 use four other codes (finite-difference and lattice-Boltzmann). Where independent codes simulate the same $(Pr, Ra)$ their $Nu$ agree to a median of 1.5% (Section 12), which is the evidence that the pooled data are homogeneous. Grid resolution differs between sources and does not need to match: what matters is that each run is converged, which every source checks (grid spacing versus Kolmogorov/Batchelor scale, points inside the thermal boundary layer, agreement of independent $Nu$ definitions).

## 1 Present paper — Shreshthi & Pandey, Table 1

Closed 2D box, $Ra = 10^9$ fixed, $Pr$ varied. All 13 rows are the paper's own flagship simulations and are included.

Table 1: Present paper, Table 1 ($Ra = 10^9$).

| Pr | Ra | Nu | Comment |
|---|---|---|---|
| 0.021 | $10^9$ | 37.43 | Included |
| 0.05 | $10^9$ | 41.51 | Included |
| 0.1 | $10^9$ | 42.72 | Included |
| 0.2 | $10^9$ | 45.98 | Included |
| 0.35 | $10^9$ | 48.1 | Included |
| 0.5 | $10^9$ | 48.64 | Included |
| 0.7 | $10^9$ | 49.22 | Included – **flag: multistable** (Sec. 13) |
| 1 | $10^9$ | 50.87 | Included |
| 2 | $10^9$ | 51.09 | Included |
| 4.38 | $10^9$ | 52.67 | Included |
| 7 | $10^9$ | 52.27 | Included |
| 20 | $10^9$ | 53.41 | Included |
| 50 | $10^9$ | 52.12 | Included |

## 2 Pandey (2021), Table 1

Closed 2D box, $Pr = 0.021$ fixed, $Ra$ varied. Runs labelled with an 'a' (e.g. 3a) are resolution checks of the preceding run — the same physical point, re-run at finer mesh, to confirm numerical convergence. Only the base run is kept; its resolution-check twin is excluded since it is not an independent data point.

Table 2: Pandey (2021), Table 1 ($Pr = 0.021$).

| Run | Ra | Nu | Comment |
|---|---|---|---|
| 1 | $5 \times 10^5$ | 5.2 | Included |
| 1a | $5 \times 10^5$ | 5.2 | Resolution check of run 1 |
| 2 | $10^6$ | 6.3 | Included |
| 2a | $10^6$ | 6.3 | Resolution check of run 2 |
| 3 | $10^7$ | 10.8 | Included |
| 3a | $10^7$ | 11.1 | Resolution check of run 3 |
| 4 | $10^8$ | 19.6 | Included |
| 5 | $10^9$ | 35.4 | Included |
| 5a | $10^9$ | 36.2 | Resolution check of run 5 |

## 3 Pandey, Tiwari & Sreenivasan (2026, in press)

### Table 1 — closed 2D box, $Pr = 0.021$ and $Pr = 0.05$

The $Ra = 10^9$ rows for both $Pr$ values report exactly the same $Nu$ as the present paper's own Table 1 (37.43 and 41.51 respectively) — the same simulation, reused across papers. Excluded here to avoid counting it twice.

Table 3: Pandey, Tiwari & Sreenivasan (2026), Table 1.

| Pr | Ra | Nu | Comment |
|---|---|---|---|
| 0.021 | $10^7$ | 10.97 | Included |
| 0.021 | $2 \times 10^7$ | 13.04 | Included |
| 0.021 | $3 \times 10^7$ | 14.42 | Included |
| 0.021 | $5 \times 10^7$ | 16.58 | Included |
| 0.021 | $10^8$ | 19.68 | Included |
| 0.021 | $4 \times 10^8$ | 28.32 | Included |
| 0.021 | $10^9$ | 37.43 | Matches Present paper exactly – duplicate |
| 0.05 | $10^7$ | 12.41 | Included |
| 0.05 | $2 \times 10^7$ | 14.52 | Included |
| 0.05 | $3 \times 10^7$ | 16.06 | Included |
| 0.05 | $10^8$ | 21.64 | Included |
| 0.05 | $3 \times 10^8$ | 29.27 | Included |
| 0.05 | $5 \times 10^8$ | 33.5 | Included |
| 0.05 | $10^9$ | 41.51 | Matches Present paper exactly – duplicate |

### Table 2 — "RBC-P", $Pr = 1$, $D = 2$ column

This table uses a different solver (DHARA, finite-difference, versus the spectral-element code used everywhere else) for a 1D/2D/3D/4D dimension-comparison study. It is a *different physical configuration* from the closed box used in every other table here: at $Pr = 1, Ra = 10^8$ it reports $Nu = 13.6$, versus $Nu = 25.01$ in Pandey & Sreenivasan (2025)'s Table 1 at the identical $Pr, Ra$ (Table 5). That gap is far too large to be numerical noise. **All rows here are excluded** from the training set.

Table 4: Pandey, Tiwari & Sreenivasan (2026), Table 2 (RBC-P, $D = 2$, $Pr = 1$).

| Ra | Nu | Comment |
|---|---|---|
| $2 \times 10^5$ | 4.8 | Different configuration (RBC-P) – excluded |
| $5 \times 10^5$ | 6.2 | Different configuration (RBC-P) – excluded |
| $10^6$ | 6.9 | Different configuration (RBC-P) – excluded |
| $2 \times 10^6$ | 8.5 | Different configuration (RBC-P) – excluded |
| $5 \times 10^6$ | 10.3 | Different configuration (RBC-P) – excluded |
| $10^7$ | 11.8 | Different configuration (RBC-P) – excluded |
| $2 \times 10^7$ | 13.1 | Different configuration (RBC-P) – excluded |
| $5 \times 10^7$ | 13.3 | Different configuration (RBC-P) – excluded |
| $10^8$ | 13.6 | Different configuration (RBC-P) – excluded |
| $2 \times 10^8$ | 15.5 | Different configuration (RBC-P) – excluded |
| $5 \times 10^8$ | 20.4 | Different configuration (RBC-P) – excluded |
| $10^9$ | 24.6 | Different configuration (RBC-P) – excluded |

## 4 Pandey & Sreenivasan (2025), Table 1

Closed 2D box, $Pr = 0.1$ and $Pr = 1$, wide $Ra$ range. As with the 2026 paper, the $Ra = 10^9$ rows duplicate the present paper's own simulations exactly (42.72 and 50.87) and are excluded.

Table 5: Pandey & Sreenivasan (2025), Table 1.

| Pr | Ra | Nu | Comment |
|---|---|---|---|
| 0.1 | $10^6$ | 5.91 | Included |
| 0.1 | $2 \times 10^6$ | 7.05 | Included |
| 0.1 | $3 \times 10^6$ | 7.88 | Included |
| 0.1 | $6 \times 10^6$ | 10.04 | Included |
| 0.1 | $10^7$ | 11.77 | Included |
| 0.1 | $2 \times 10^7$ | 15.35 | Included |
| 0.1 | $3 \times 10^7$ | 17.5 | Included |
| 0.1 | $10^8$ | 23.01 | Included |
| 0.1 | $3 \times 10^8$ | 31.06 | Included |
| 0.1 | $10^9$ | 42.72 | Matches Present paper exactly – duplicate |
| 0.1 | $3 \times 10^9$ | 59.15 | Included |
| 0.1 | $10^{10}$ | 84.74 | Included |
| 1 | $3 \times 10^6$ | 6.01 | Included |
| 1 | $6 \times 10^6$ | 7.11 | Included |

| Pr | Ra | Nu | Comment |
|---|---|---|---|
| 1 | $8 \times 10^6$ | 8.18 | Included |
| 1 | $9 \times 10^6$ | 8.64 | Included |
| 1 | $10^7$ | 9.03 | Included |
| 1 | $1.5 \times 10^7$ | 13.33 | Included |
| 1 | $2 \times 10^7$ | 14.75 | Included |
| 1 | $3 \times 10^7$ | 17.27 | Included |
| 1 | $6 \times 10^7$ | 21.48 | Included |
| 1 | $10^8$ | 25.01 | Included |
| 1 | $3 \times 10^8$ | 35.61 | Included |
| 1 | $6 \times 10^8$ | 44.15 | Included |
| 1 | $10^9$ | 50.87 | Matches Present paper exactly – duplicate |
| 1 | $3 \times 10^9$ | 67.51 | Included |
| 1 | $10^{10}$ | 94.52 | Included |
| 1 | $3 \times 10^{10}$ | 130.2 | Included |
| 1 | $10^{11}$ | 183.3 | Included |
| 1 | $3 \times 10^{11}$ | 262.3 | Included |
| 1 | $10^{12}$ | 394.2 | Included |

## 5 van der Poel, Stevens & Lohse (2013), Appendix Table 2

2D, $\Gamma = 1$, no-slip on all walls, $Ra = 10^8$, $Pr$ varied (2D code of Sugiyama et al. 2009). The text does not restate the sidewall thermal condition, but van der Poel et al. (2012), using the same code, state adiabatic sidewalls explicitly (Section 10), and the $Pr = 1$ value (25.17) matches Pandey & Sreenivasan (2025) (25.01) to 0.6%. Rows in bold in the original are resolution checks: as in Section 2, only one run per physical point is kept, here the finest grid. The value used is $Nu_f$ (whole run); $Nu_h$ (last half) indicates time convergence. Not used: the 3D table (Table 1), Fig. 5(b) ($\Gamma = 0.5$) and the $Ra = 10^6$ series in Fig. 6(b) (partly double-roll states).

Table 6: van der Poel et al. (2013), Table 2 (2D, $Ra = 10^8$).

| Pr | Grid | $N_{BL}$ | $N_{BL}^{min}$ | $h$ | $Nu_f$ | $Nu_h$ | $\tau_f$ | Comment |
|---|---|---|---|---|---|---|---|---|
| 55 | 512×256 | 16 | 7 | 1.27 | 27.83 | 27.86 | 50000 | Included |
| 55 | 256×128 | 8 | 7 | 2.38 | 27.76 | 27.75 | 50000 | Coarser-grid resolution check – excluded |
| 30 | 256×128 | 8 | 7 | 2.38 | 27.50 | 27.51 | 40000 | Included |
| 20 | 256×128 | 8 | 7 | 2.38 | 27.44 | 27.44 | 30000 | Included |
| 10 | 256×128 | 8 | 7 | 2.37 | 27.25 | 27.26 | 20000 | Included |
| 6.4 | 512×256 | 16 | 7 | 1.26 | 26.99 | 26.99 | 10000 | Included |
| 6.4 | 256×128 | 8 | 7 | 2.36 | 26.79 | 26.79 | 20000 | Coarser-grid resolution check – excluded |
| 4.38 | 256×128 | 8 | 7 | 2.33 | 25.51 | 25.47 | 30000 | Included |
| 3 | 256×128 | 8 | 7 | 2.32 | 25.07 | 25.05 | 10000 | Included |
| 1.5 | 256×128 | 8 | 7 | 2.31 | 24.83 | 24.82 | 10000 | Included |
| 1 | 512×256 | 8 | 7 | 2.32 | 25.17 | 25.16 | 10000 | Included (NBL/h cells look copied from the row above) |

| **Pr** | **Grid** | $N_{BL}$ | $N_{BL}^{min}$ | $h$ | $Nu_f$ | $Nu_h$ | $\tau_f$ | **Comment** |
|---|---|---|---|---|---|---|---|---|
| 0.7 | 1024×512 | 30 | 8 | 0.78 | 25.22 | 25.10 | 8000 | Included |
| 0.7 | 512×256 | 15 | 8 | 1.48 | 25.14 | 25.16 | 8000 | Coarser-grid resolution check – excluded |
| 0.45 | 1024×512 | 34 | 11 | 0.96 | 24.78 | 24.59 | 2000 | Included |
| 0.3 | 1024×512 | 37 | 13 | 1.18 | 25.25 | 25.09 | 2000 | Included |
| 0.2 | 1024×512 | 34 | 17 | 1.43 | 25.73 | 25.40 | 1500 | Included |
| 0.1 | 2048×1024 | 76 | 26 | 1.01 | 23.63 | 23.18 | 200 | Included |
| 0.1 | 1024×512 | 36 | 26 | 1.97 | 23.32 | 23.03 | 1000 | Coarser-grid resolution check – excluded |
| 0.065 | 2048×1024 | 82 | 34 | 1.23 | 21.76 | 22.07 | 200 | Included – short average ($200\,t_f$) |
| 0.045 | 2048×1024 | 102 | 43 | 1.46 | 20.52 | 20.95 | 80 | Included – short average ($80\,t_f$) |

## 6 Zhang & Zhou (2024), Tables I and II

2D square box, no-slip on all four walls, isothermal plates, insulated sidewalls — identical configuration. Compact fourth-order finite differences. Every run has $\Delta_g/\eta < 1$ and meets the Shishkina et al. boundary-layer criterion, and the three independent $Nu$ estimates agree within 3%. All 15 rows are included. This source extends the $Pr$ range down to 0.005 (previous minimum 0.021).

Table 7: Zhang & Zhou (2024), Tables I–II.

| **Pr** | **Ra** | **Grid** | $\Delta_g/\eta$ | $N_{BL}$ | $t_{avg}$ | $Nu$ | $Nu_{\varepsilon_u}$ | $Nu_{\varepsilon_\theta}$ | **Comment** |
|---|---|---|---|---|---|---|---|---|---|
| 0.005 | $10^7$ | 1537×1537 | 0.87 | 86 | 105 | 8.91 | 8.65 | 8.91 | Included – short average ($105\,t_f$) |
| 0.007 | $10^7$ | 1537×1537 | 0.74 | 83 | 130 | 9.25 | 9.01 | 9.25 | Included – short average ($130\,t_f$) |
| 0.01 | $10^7$ | 1025×1025 | 0.95 | 52 | 325 | 9.89 | 9.69 | 9.89 | Included |
| 0.015 | $10^7$ | 1025×1025 | 0.79 | 49 | 387 | 10.49 | 10.45 | 10.49 | Included |
| 0.023 | $10^7$ | 769×769 | 0.86 | 35 | 684 | 11.07 | 11.10 | 11.09 | Included |
| 0.035 | $10^7$ | 769×769 | 0.71 | 32 | 1035 | 11.86 | 11.87 | 11.86 | Included |
| 0.05 | $10^7$ | 513×513 | 0.90 | 21 | 2400 | 12.32 | 12.32 | 12.34 | Included |
| 0.07 | $10^7$ | 513×513 | 0.77 | 20 | 3600 | 12.96 | 12.99 | 12.98 | Included |
| 0.01 | $10^8$ | 2049×2049 | 0.98 | 57 | 140 | 18.06 | 17.53 | 18.11 | Included – **flag: conflict** (Sec. 13) |
| 0.023 | $10^8$ | 1537×1537 | 0.88 | 38 | 360 | 20.07 | 19.61 | 20.01 | Included |
| 0.035 | $10^8$ | 1537×1537 | 0.73 | 37 | 416 | 20.78 | 20.82 | 20.81 | Included |
| 0.05 | $10^8$ | 1025×1025 | 0.93 | 24 | 1000 | 21.71 | 21.56 | 21.78 | Included |
| 0.07 | $10^8$ | 1025×1025 | 0.79 | 23 | 800 | 22.32 | 22.20 | 22.33 | Included |
| 0.1 | $10^8$ | 769×769 | 0.89 | 17 | 2400 | 23.21 | 23.19 | 23.19 | Included |
| 0.15 | $10^8$ | 769×769 | 0.74 | 16 | 2000 | 24.18 | 24.22 | 24.27 | Included |

## 7 Zhang, Zhou & Sun (2017), Table 1

2D box of unit aspect ratio, adiabatic sidewalls, no-slip on all surfaces; same compact fourth-order code as Zhang & Zhou (2024). Columns 5–6 compare the measured dissipation rates with the exact relations (1 = perfect); nearly all are within 1%. All 18 rows are included.

Table 8: Zhang, Zhou & Sun (2017), Table 1.

| **Pr** | **Ra** | **Grid** | $Nu$ | $\varepsilon_u$ ratio | $\varepsilon_\theta$ ratio | $N_{BL}$ | **Comment** |
|---|---|---|---|---|---|---|---|
| 0.7 | $10^6$ | 129×129 | 6.3 | 1.0041 | 1.0023 | 11 | Included |
| 0.7 | $3\times10^6$ | 193×193 | 7.64 | 1.0038 | 1.0018 | 13 | Included |
| 0.7 | $10^7$ | 257×257 | 11.37 | 1.0031 | 1.0014 | 12 | Included |
| 0.7 | $3\times10^7$ | 385×385 | 16.5 | 1.0025 | 1.0006 | 12 | Included |
| 0.7 | $10^8$ | 513×513 | 25.25 | 1.0021 | 0.9998 | 11 | Included |
| 0.7 | $3\times10^8$ | 769×769 | 35.73 | 1.0017 | 1.0006 | 11 | Included |
| 0.7 | $10^9$ | 1025×1025 | 53.51 | 0.9967 | 0.9996 | 10 | Included – **flag: multistable** (Sec. 13) |
| 0.7 | $3\times10^9$ | 1537×1537 | 66.95 | 0.9896 | 1.0013 | 12 | Included |
| 0.7 | $10^{10}$ | 2049×2049 | 93.88 | 1.0358 | 0.9996 | 12 | Included – $\varepsilon_u$ check off by 3.6% |
| 5.3 | $10^6$ | 129×129 | 6.87 | 1.0026 | 1.0019 | 10 | Included |
| 5.3 | $3\times10^6$ | 193×193 | 9.33 | 1.0022 | 1.0009 | 11 | Included |
| 5.3 | $10^7$ | 257×257 | 13.28 | 1.0017 | 1.0008 | 10 | Included |
| 5.3 | $3\times10^7$ | 385×385 | 18.86 | 1.0010 | 1.0009 | 11 | Included |
| 5.3 | $10^8$ | 513×513 | 26.21 | 1.0000 | 1.0002 | 10 | Included |
| 5.3 | $3\times10^8$ | 769×769 | 35.96 | 1.0016 | 1.0001 | 11 | Included |
| 5.3 | $10^9$ | 1025×1025 | 51.28 | 1.0030 | 1.0005 | 11 | Included |
| 5.3 | $3\times10^9$ | 1537×1537 | 71.57 | 0.9996 | 1.0019 | 11 | Included |
| 5.3 | $10^{10}$ | 3073×3073 | 103 | 0.9963 | 1.0003 | 15 | Included |

## 8 Xu, Shi & Xi (2019), Tables I and II

2D square cell ($L = H$), no-slip on all four walls, adiabatic vertical walls, isothermal plates; lattice-Boltzmann (D2Q9/D2Q5). $\max(\Delta_g/\eta, \Delta_g/\eta_B) \le 0.45$ and the three $Nu$ definitions agree to about 1%. The volume-averaged $Nu_{vol}$ is used. All 8 rows are included. At $Ra = 10^6$ it gives 6.28, against 6.3 from Pandey (2021) at the nearby $Pr = 0.021$.

Table 9: Xu, Shi & Xi (2019), Tables I–II.

| **Pr** | **Ra** | **Grid** | $\Delta_g/\eta$ | $t_{avg}$ | $Nu_{vol}$ | $Nu_{wall}$ | $Nu_{th}$ | **Comment** |
|---|---|---|---|---|---|---|---|---|
| 0.025 | $10^6$ | 769×769 | 0.39 | 700 | 6.28 | 6.25 | 6.35 | Included |
| 0.025 | $1.3\times10^6$ | 851×851 | 0.39 | 2300 | 6.92 | 6.97 | 7.01 | Included |
| 0.025 | $1.6\times10^6$ | 901×901 | 0.40 | 1400 | 7.37 | 7.36 | 7.32 | Included |
| 0.025 | $2\times10^6$ | 1001×1001 | 0.38 | 1000 | 7.72 | 7.74 | 7.73 | Included |
| 0.025 | $3\times10^6$ | 1025×1025 | 0.43 | 1400 | 8.54 | 8.57 | 8.57 | Included |
| 0.025 | $4\times10^6$ | 1101×1101 | 0.43 | 1000 | 9.08 | 9.13 | 9.14 | Included |
| 0.025 | $6\times10^6$ | 1201×1201 | 0.45 | 1300 | 9.93 | 9.99 | 10.00 | Included |
| 0.025 | $10^7$ | 1537×1537 | 0.42 | 1000 | 11.38 | 11.39 | 11.42 | Included |

## 9 He, Fang, Gao, Huang & Bao (2021), Fig. 5(a)

2D square box ($\Gamma = 1$), no-slip velocity on all walls, isothermal plates, adiabatic sidewalls; second-order finite differences, meshes set by the Gao et al. criteria, $Nu$ from different methods within 1%, averaged over 200–500 free-fall times. Ten $Pr$ values from 0.25 to 100, $Ra$ from $10^7$ to $10^{10}$ (low $Pr$ is not run at the highest $Ra$). Periods of large-scale-flow reversal are left out of their $Nu$ averages; the authors report this changes $Nu$ by at most 3%.

$Nu$ is not tabulated. The values below were extracted from the vector markers of Fig. 5(a) ($Nu/Ra^{0.3}$ on a linear axis), as described in Section 14; estimated error below 0.2%. All 62 points are included.

Table 10: He et al. (2021), values extracted from Fig. 5(a).

| **Pr** | **Ra** | $Nu/Ra^{0.3}$ | $Nu$ | **Comment** |
|---|---|---|---|---|
| 0.25 | $10^7$ | 0.0885 | 11.14 | Included |
| 0.25 | $2 \times 10^7$ | 0.0960 | 14.87 | Included |
| 0.25 | $5 \times 10^7$ | 0.1000 | 20.40 | Included |
| 0.25 | $10^8$ | 0.1012 | 25.41 | Included |
| 0.4 | $10^7$ | 0.0915 | 11.51 | Included |
| 0.4 | $2 \times 10^7$ | 0.0931 | 14.43 | Included |
| 0.4 | $5 \times 10^7$ | 0.0990 | 20.19 | Included – two overplotted markers averaged |
| 0.4 | $10^8$ | 0.0977 | 24.55 | Included – two overplotted markers averaged |
| 0.4 | $2 \times 10^8$ | 0.1028 | 31.78 | Included |
| 0.7 | $10^7$ | 0.0906 | 11.40 | Included |
| 0.7 | $2 \times 10^7$ | 0.0926 | 14.35 | Included |
| 0.7 | $5 \times 10^7$ | 0.0967 | 19.74 | Included |
| 0.7 | $10^8$ | 0.1006 | 25.26 | Included |
| 0.7 | $2 \times 10^8$ | 0.1012 | 31.29 | Included |
| 0.7 | $5 \times 10^8$ | 0.1037 | 42.20 | Included |
| 1.2 | $2 \times 10^7$ | 0.0962 | 14.90 | Included |
| 1.2 | $5 \times 10^7$ | 0.1011 | 20.64 | Included |
| 1.2 | $10^8$ | 0.1003 | 25.19 | Included |
| 1.2 | $2 \times 10^8$ | 0.1013 | 31.33 | Included |
| 1.2 | $5 \times 10^8$ | 0.1017 | 41.38 | Included |
| 1.2 | $10^9$ | 0.1024 | 51.33 | Included |
| 2 | $10^7$ | 0.0956 | 12.03 | Included |
| 2 | $2 \times 10^7$ | 0.0978 | 15.15 | Included |
| 2 | $5 \times 10^7$ | 0.1003 | 20.46 | Included |
| 2 | $10^8$ | 0.0997 | 25.04 | Included |
| 2 | $2 \times 10^8$ | 0.1002 | 30.99 | Included |
| 2 | $5 \times 10^8$ | 0.1003 | 40.85 | Included |
| 2 | $10^9$ | 0.0999 | 50.07 | Included |
| 2 | $2 \times 10^9$ | 0.1020 | 62.92 | Included |
| 4.3 | $10^7$ | 0.1035 | 13.03 | Included |
| 4.3 | $2 \times 10^7$ | 0.1059 | 16.42 | Included |
| 4.3 | $5 \times 10^7$ | 0.1047 | 21.36 | Included |
| 4.3 | $10^8$ | 0.1034 | 25.96 | Included |
| 4.3 | $2 \times 10^8$ | 0.1022 | 31.60 | Included |
| 4.3 | $5 \times 10^8$ | 0.1018 | 41.44 | Included |

| Pr | Ra | $Nu/Ra^{0.3}$ | $Nu$ | Comment |
|---|---|---|---|---|
| 4.3 | $10^9$ | 0.1014 | 50.82 | Included |
| 4.3 | $2\times10^9$ | 0.1024 | 63.16 | Included |
| 4.3 | $5\times10^9$ | 0.1037 | 84.19 | Included |
| 10 | $2\times10^8$ | 0.1083 | 33.51 | Included |
| 10 | $5\times10^8$ | 0.1065 | 43.35 | Included |
| 10 | $10^9$ | 0.1053 | 52.76 | Included |
| 10 | $2\times10^9$ | 0.1051 | 64.83 | Included |
| 10 | $5\times10^9$ | 0.1045 | 84.89 | Included |
| 10 | $10^{10}$ | 0.1051 | 105.14 | Included |
| 20 | $2\times10^8$ | 0.1097 | 33.93 | Included |
| 20 | $5\times10^8$ | 0.1085 | 44.15 | Included |
| 20 | $10^9$ | 0.1072 | 53.71 | Included |
| 20 | $2\times10^9$ | 0.1065 | 65.72 | Included |
| 20 | $5\times10^9$ | 0.1058 | 85.95 | Included |
| 20 | $10^{10}$ | 0.1059 | 105.85 | Included |
| 50 | $2\times10^8$ | 0.1098 | 33.97 | Included |
| 50 | $5\times10^8$ | 0.1081 | 44.00 | Included |
| 50 | $10^9$ | 0.1068 | 53.54 | Included |
| 50 | $2\times10^9$ | 0.1065 | 65.72 | Included |
| 50 | $5\times10^9$ | 0.1057 | 85.88 | Included |
| 50 | $10^{10}$ | 0.1061 | 106.08 | Included |
| 100 | $2\times10^8$ | 0.1112 | 34.38 | Included |
| 100 | $5\times10^8$ | 0.1104 | 44.94 | Included |
| 100 | $10^9$ | 0.1085 | 54.40 | Included |
| 100 | $2\times10^9$ | 0.1071 | 66.07 | Included |
| 100 | $5\times10^9$ | 0.1060 | 86.07 | Included |
| 100 | $10^{10}$ | 0.1053 | 105.34 | Included |

## 10 van der Poel, Stevens, Sugiyama & Lohse (2012), Fig. 1(a)

2D, no-slip on all walls, isothermal plates, adiabatic sidewalls, same fourth-order finite-difference code as Section 5. Only the $\Gamma = 1$ series (purple pentagrams, $Pr = 1$) is used; the other aspect ratios are a different configuration. $Nu$ is not tabulated; values were extracted from the vector markers (Section 14). The $Ra = 10^8$ point is the same simulation as van der Poel (2013) and is excluded.

Table 11: van der Poel et al. (2012), $\Gamma = 1$ values extracted from Fig. 1(a).

| Pr | Ra | $Nu$ | Comment |
|---|---|---|---|
| 1 | $10^7$ | 8.47 | Included – **flag: flow-state** (Sec. 13) |
| 1 | $2\times10^7$ | 13.76 | Included – **flag: flow-state** (Sec. 13) |
| 1 | $4\times10^7$ | 18.58 | Included |
| 1 | $7\times10^7$ | 22.44 | Included |
| 1 | $10^8$ | 25.14 | Same run as van der Poel (2013) Table 2 (25.17) – duplicate |
| 1 | $2\times10^8$ | 31.52 | Included |

| Pr | Ra | $Nu$ | Comment |
|---|---|---|---|
| 1 | $3 \times 10^8$ | 35.83 | Included |
| 1 | $4 \times 10^8$ | 38.94 | Included |
| 1 | $6 \times 10^8$ | 43.96 | Included |
| 1 | $8 \times 10^8$ | 47.85 | Included |
| 1 | $10^9$ | 51.87 | Included |
| 1 | $2 \times 10^9$ | 61.37 | Included |
| 1 | $3 \times 10^9$ | 68.46 | Included |
| 1 | $4 \times 10^9$ | 73.15 | Included |
| 1 | $6 \times 10^9$ | 82.96 | Included |
| 1 | $8 \times 10^9$ | 89.60 | Included |
| 1 | $10^{10}$ | 96.97 | Included |

## 11 Yang et al. (2021): checked, not used

Same configuration (2D square box, no-slip, insulated sidewalls), but the paper studies rough plates. Its smooth-cell reference values appear only in Fig. 2(a), a low-resolution raster image (about 200 pixels wide) at $Ra = 10^8$, which is already the best-covered Rayleigh number in the dataset, so digitizing it would add error without adding coverage. The text gives one smooth-cell value, $Nu = 50.85$ at $Pr = 1$, $Ra = 10^9$, which agrees with the present paper (50.87) and is not added. The figure is still useful as a check: at $Pr = 0.01$, $Ra = 10^8$ it reads $Nu \approx 16.3$, about 10% below Zhang & Zhou (2024), which is why that row is flagged.

## 12 Cross-source agreement

Points simulated by more than one source (same $Pr$ and $Ra$). Spread is (max − min)/mean. The median spread is 1.5%. Pairs within one code family (e.g. Pandey 2021 vs. Pandey, Tiwari & Sreenivasan 2026) and pairs across independent codes agree equally well, except at the flagged points (red), which are explained in Section 13. Abbreviations: Present = Shreshthi & Pandey; P21, PTS26, PS25 = refs. 2–4; vdP13, ZZ24, ZZS17, XSX19, He21, vdP12 = refs. 5–10.

Table 12: Points reported by more than one source.

| Pr | Ra | Values | Spread (%) |
|---|---|---|---|
| 0.7 | $10^8$ | vdP13 25.22; ZZS17 25.25; He21 25.26 | 0.16 |
| 0.7 | $10^7$ | ZZS17 11.37; He21 11.4 | 0.26 |
| 0.05 | $10^8$ | PTS26 21.64; ZZ24 21.71 | 0.32 |
| 0.021 | $10^8$ | P21 19.6; PTS26 19.68 | 0.41 |
| 1 | $6 \times 10^8$ | PS25 44.15; vdP12 43.96 | 0.43 |
| 20 | $10^9$ | Present 53.41; He21 53.71 | 0.56 |
| 1 | $3 \times 10^8$ | PS25 35.61; vdP12 35.83 | 0.62 |
| 1 | $10^8$ | PS25 25.01; vdP13 25.17 | 0.64 |
| 0.05 | $10^7$ | PTS26 12.41; ZZ24 12.32 | 0.73 |
| 1 | $3 \times 10^9$ | PS25 67.51; vdP12 68.46 | 1.40 |
| 0.021 | $10^7$ | P21 10.8; PTS26 10.97 | 1.56 |
| 1 | $10^9$ | Present 50.87; vdP12 51.87 | 1.95 |

| Pr | Ra | Values | Spread (%) |
|---|---|---|---|
| 2 | $10^9$ | Present 51.09; He21 50.07 | 2.02 |
| 1 | $10^{10}$ | PS25 94.52; vdP12 96.97 | 2.56 |
| 0.1 | $10^8$ | PS25 23.01; ZZ24 23.21; vdP13 23.63 | 2.66 |
| 50 | $10^9$ | Present 52.12; He21 53.54 | 2.69 |
| 0.021 | $10^9$ | Present 37.43; P21 35.4 | 5.57 |
| 1 | $10^7$ | PS25 9.03; vdP12 8.47 | 6.40 |
| 1 | $2 \times 10^7$ | PS25 14.75; vdP12 13.76 | 6.94 |
| 0.7 | $10^9$ | Present 49.22; ZZS17 53.51 | 8.35 |

## 13 Flagged rows

These rows are kept in `nu_dataset_model.csv` and marked in the `Flag` column of `nu_dataset_sources.csv`, so a model can be re-run without them.

- **multistable** ($Pr = 0.7$, $Ra = 10^9$): the present paper has 49.22, Zhang, Zhou & Sun (2017) 53.51. van der Poel et al. (2012, Sec. V) show that at exactly this point with $\Gamma \approx 1$ the flow can sit in a condensed single-roll state ($Nu \approx 53$) or an uncondensed state ($Nu \approx 49$–$50$), each stable for thousands of turnover times. Both values are physically correct; for a model that only sees $(Pr, Ra)$ the difference is irreducible scatter.
- **flow-state** ($Pr = 1$, $Ra \leq 2 \times 10^7$, van der Poel 2012): 6–7% below Pandey & Sreenivasan (2025), whose paper locates the steady/transient transition in this range. The two runs most likely sit in different flow states.
- **conflict** ($Pr = 0.01$, $Ra = 10^8$, Zhang & Zhou 2024): 18.06, while their own dissipation-based estimate is 17.53 and Yang et al. (2021) read about 16.3. Averaged over only 140 free-fall times.

Table 13: Flagged rows.

| Pr | Ra | $Nu$ | Source | Flag |
|---|---|---|---|---|
| 0.7 | $10^9$ | 49.22 | Present | multistable |
| 0.01 | $10^8$ | 18.06 | ZZ24 | conflict |
| 0.7 | $10^9$ | 53.51 | ZZS17 | multistable |
| 1 | $10^7$ | 8.47 | vdP12 | flow-state |
| 1 | $2 \times 10^7$ | 13.76 | vdP12 | flow-state |

## 14 Extracting values from figures

For He et al. (2021) and van der Poel et al. (2012) the plots are stored in the PDF as vector graphics, so the markers' exact coordinates can be read from the drawing commands instead of estimated from a picture (PyMuPDF `get_drawings`):

1. Axis calibration from the tick marks and tick labels drawn in the same figure (e.g. He Fig. 5(a): 0.01 in $Nu/Ra^{0.3}$ per 42.01 pt; 48.27 pt per decade of $Ra$).

2. Each marker identified by its fill or stroke colour and shape, which map one-to-one to the legend; legend entries removed by position. Open markers drawn as one combined path were split by clustering vertices.

3. Marker centre = bounding-box centre (squares, circles, diamonds, triangles) or vertex mean (regular pentagons, i.e. the circumcentre).

4. Every marker lands within 0.002 decades of a nominal $Ra$ value, confirming the calibration. $Nu$ then follows from $Nu/Ra^{0.3}$ times $Ra^{0.3}$. Two He markers drawn on top of each other were averaged (they differ by 0.3%).

The positional uncertainty (about 0.1 pt) corresponds to under 0.2% in $Nu$. Raster figures (Yang et al. 2021, He et al. Fig. 5(b)) were not digitized.

## 15 Summary

Table 14: Row counts by source.

| Source | Rows in source | Included | Excluded |
|---|---|---|---|
| Present paper, Table 1 | 13 | 13 | 0 |
| Pandey (2021), Table 1 | 9 | 5 | 4 |
| Pandey, Tiwari & Sreenivasan (2026), Table 1 | 14 | 12 | 2 |
| Pandey, Tiwari & Sreenivasan (2026), Table 2 (RBC-P) | 12 | 0 | 12 |
| Pandey & Sreenivasan (2025), Table 1 | 31 | 29 | 2 |
| van der Poel et al. (2013), Table 2 (2D) | 20 | 16 | 4 |
| Zhang & Zhou (2024), Table II | 15 | 15 | 0 |
| Zhang, Zhou & Sun (2017), Table 1 | 18 | 18 | 0 |
| Xu, Shi & Xi (2019), Table II | 8 | 8 | 0 |
| He et al. (2021), Fig. 5(a) (extracted) | 62 | 62 | 0 |
| van der Poel et al. (2012), Fig. 1(a), $\Gamma = 1$ (extracted) | 17 | 16 | 1 |
| **Total** | **219** | **194** | **25** |

The 194 included rows make up `data/nu_dataset_model.csv` (columns `Pr, Ra, Nu`); `data/nu_dataset_sources.csv` holds the same rows with `Source`, `Grid`, `Notes` and `Flag` columns. They span $Pr$ from 0.005 to 100, $Ra$ from $5 \times 10^5$ to $10^{12}$, $Nu$ from 5.2 to 394.2, at 172 distinct $(Pr, Ra)$ points.

**Coverage gaps.** Above $Ra = 10^{10}$ the only data are the $Pr = 1$ series of Pandey & Sreenivasan (2025), so any prediction there at other $Pr$ is extrapolation. Below $Pr = 0.02$ only $Ra = 10^7$ and $10^8$ are covered (Zhang & Zhou 2024).

**Adding further sources.** Follow the same process: reproduce the table, confirm the configuration (closed $\Gamma = 1$ box, no-slip, adiabatic sidewalls) before including any row, check agreement at overlapping points, and update the summary above. Candidates: Huang & Zhou (2013, J. Fluid Mech.) and the exact table behind He et al. (2021), both obtainable from the authors.